Projecte Intermodular – UT0

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1. Project title

Online vinyl record store with crypto payment

2. Introduction and context.

Vinyl has experienced a significant revival over the last 15-20 years. This resurgence is driven not only by retro aesthetics but by a growing fatigue with the "chronically online" lifestyle. Fatigued by constant multitasking and delegating one's music choice to algorithms, people seek intentionality when doing basic tasks as well as listening to music. Vinyl helps to disconnect from other things and focus the music through physical touch, reading additional information on the sleeve, looking at the artwork which is much bigger than it appears on the smartphone screen and having to manually load a side and then change it.

Usually online record shops use web 2.0 banking systems like Visa and Mastercard and rarely offer payment by crypto. This website can be marketed as one for crypto enthusiasts.

1) Enabling traditional payment systems for a student project can prove to be too difficult because it involves complex bureaucracy and compliance standards.
2) Enabling payment by crypto is an achievable task and will look great in my portfolio.

3. Objectives.

- General objective: To design and develop a web application for vinyl record store with payment in crypto.
- Technical objective: to develop a scalable full-stack application using MERN stack (MongoDB, Express.js, React, Node.js).
- Functional objectives:
    1) Browse the available stock
    2) Preview artwork and audio
    3) Register user to save billing and delivery info
    4) Payment with crypto.

4. Impact and ethical aspects.

1) Accessibility implementation:

a) Screen readers - add special tags to web elements so people with vision impairments can read them with text-to-speech software.

b) Colors should have high contrast to ease distinction between different elements and text.

2) Usability:

a) Navigation should be simple
b) Simplify and streamline the buying process

3) Data protection

Crypto is a big advantage in terms of data protection.

4) Software licenses

The MERN stack is free and open source. The advantages are zero cost in licensing and huge community support by other enthusiasts.

Component	**Technology Type**	**Primary License**	**Licensing Obligations**
MongoDB	Document Database	**Server Side Public License (SSPL)** (Community Edition)	**Strong Copyleft (Viral)**. If you offer the software or a modified version as a service (SaaS, cloud service), you must make the entire application source code, including all management software, freely available under the SSPL. For internal use or if not offered as a service, traditional use is less restricted.
Express.js	Node.js Web Framework	**MIT License**	**Permissive**. Allows almost any use, including proprietary, commercial, and redistribution. Requires that the original copyright and license notice are included in all copies or substantial portions of the software.
React	JavaScript UI Library	**MIT License**	**Permissive**. Allows almost any use, including proprietary, commercial, and redistribution. Requires that the original copyright and license notice are included in all copies or substantial portions of the software.

Node.js	JavaScript Runtime	**MIT License**	**Permissive**. Allows almost any use, including proprietary, commercial, and redistribution. Requires that the original copyright and license notice are included in all copies or substantial portions of the software.

5) Diversity and inclusion:

The main unifying factor for the buyers will be payment by crypto, so to maximize the number of clients:

1) The store should sell music of different genres.
2) The interface should be neutral in register and convey only factual information.

5. Methodology and Tools

Methodology: Agile approach, specifically the **Kanban** method to manage tasks and workflow.

Project Management: Trello will be used to track progress with a board divided into "Backlog," "Sprint 1," "Doing," and "Done."

Development Stack (MERN):

Database: MongoDB (NoSQL database for storing album metadata and user info).

Backend: Node.js with **Express.js** (to build the RESTful API).

Frontend: React.js (to create a dynamic Single Page Application).

Version Control: Git and **GitHub** (for code hosting and version history).

Design & Additional Tools:

UI/UX Design: Figma (for creating wireframes and prototypes).

Payments: Ethereum Blockchain (integration via Web3 libraries/MetaMask) for decentralized transactions.

IDE: Visual Studio Code (primary code editor).

6. Planning

Phase	Description	Dates	Expected Results (Deliverables)
Phase 1	**Requirements definition and market analysis**	Nov 3 – Nov 20	Requirements document and analysis

Phase 2	**UI Design and Database**	**Jan 2 – Jan 25**	Figma prototype and ER diagram
Phase 3	**Base architecture development (backend/frontend)**	**Jan 29 – Feb 18**	Minimum Viable Prototype (MVP)
Phase 4	**Cart, payments, and authentication implementation**	**Feb 19 – Apr 19**	Version with core features
Phase 5	**Optimization, testing, and responsive design**	**Apr 22 – Jun 15**	Stable and complete version
Final Delivery	**Documentation preparation and presentation**	**By Jun 30**	First complete version of the project

Links

https://en.wikipedia.org/wiki/Vinyl_revival